

Instructions:

1. Run task_0a.py to generate the vector database if not already generated.
2. Press Run All (or restart kernel and run all cells).
3. You will be prompted to provide input values.

– Task 1: Implement a program which, given (a) a query label, l (b) a user selected feature space, and (c) positive integer k, identifies and visualizes the most relevant k images for the given label l, along with their scores, under the selected feature space.

```
In [ ]: LABEL = input("Provide a valid label name / label ID present in the Caltech101 dataset")
FEATURE_SPACE = input("Provide a feature space [color, hog, avgpool, layer3, layer4, layer5, layer5_pool, layer5_pool_512x512]")
K = int(input("Enter K, the number of similar images K to find for the selected label and feature space"))
```

```
In [ ]: # Find the k most similar images for a given query label for the given feature space
from utils.query_input_processor import check_label_is_valid

LABEL = check_label_is_valid(LABEL)
```

Input label: 100
 Downloading dataset if not present.
 Files already downloaded and verified
 Output label: yin_yang

```
In [ ]: # Load feature space and representative label vectors for that feature space
from utils.database_utils import retrieve

feature_vectors = retrieve(f'{FEATURE_SPACE}.pt')
rep_label_vectors = retrieve(f'rep_label_{FEATURE_SPACE}.pt')
```

```
In [ ]: from utils.dataset_utils import initialize_dataset
dataset = initialize_dataset()

from utils.distance_utils import get_distance_fn, top_k_min_distance_ranker
top_k = top_k_min_distance_ranker(K, rep_label_vectors[LABEL], feature_vectors)

print("\nK =", K, "most similar images by", FEATURE_SPACE, "feature space using", dataset.feature_space)
```

```
for i in range(len(top_k)):

    distance, image_id, label = top_k[i]

    print(str(i + 1) + ".", "\tImage ID: ", image_id, "\t\tDistance: ", distance)
    display(dataset[image_id][0])
```

Downloading dataset if not present.
Files already downloaded and verified

K = 10 most similar images by layer3 feature space using distance: cosine

1. Image ID: 8672 Distance: 0



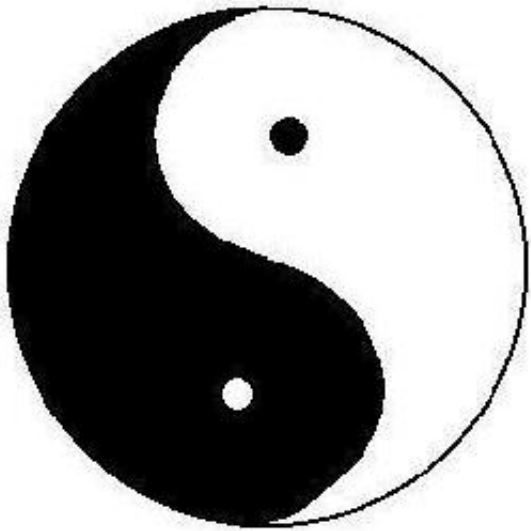
2. Image ID: 8618

Distance: 0.0054267048835754395



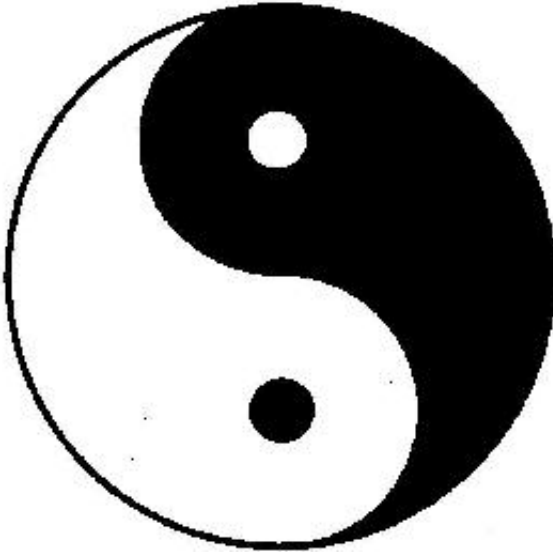
3. Image ID: 8620

Distance: 0.006788074970245361



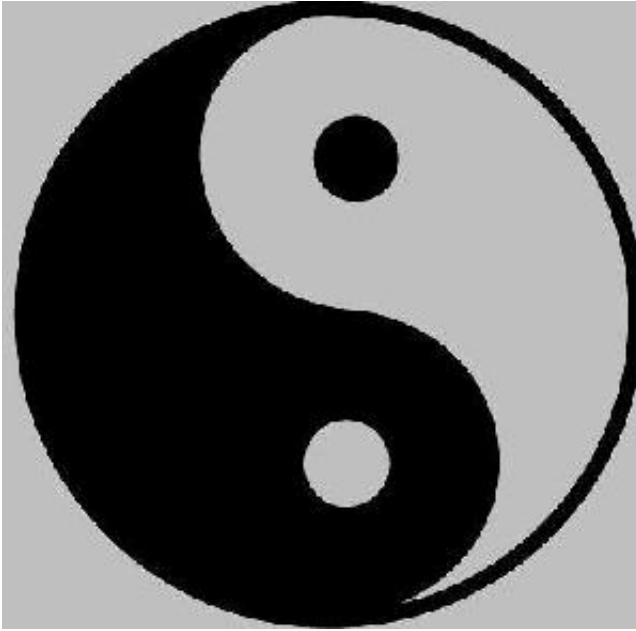
4. Image ID: 8656

Distance: 0.006886482238769531



5. Image ID: 8630

Distance: 0.009424269199371338



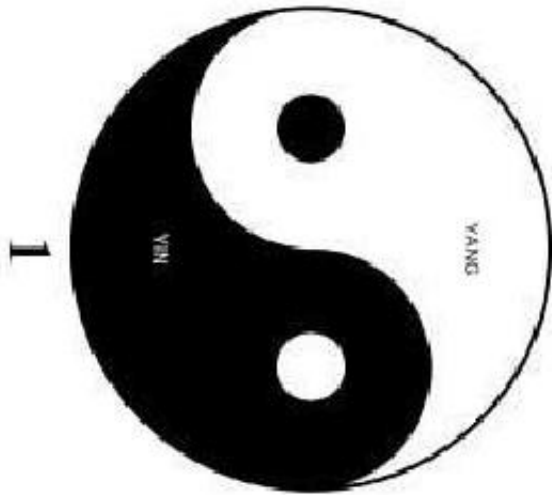
6. Image ID: 8654

Distance: 0.009434163570404053



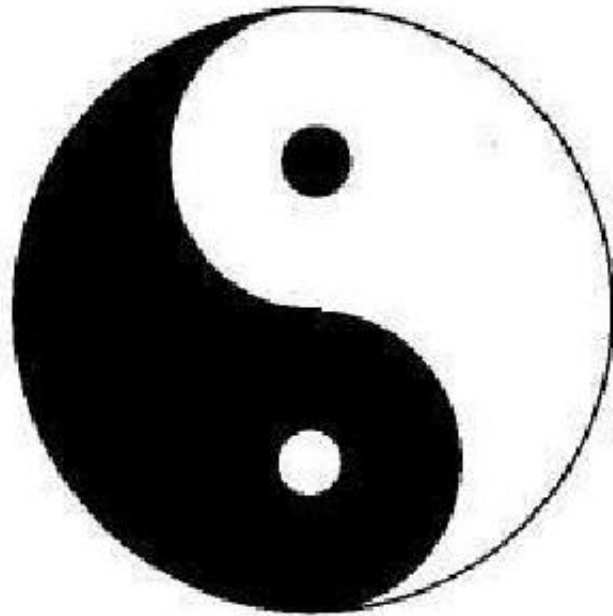
7. Image ID: 8666

Distance: 0.009660303592681885



8. Image ID: 8658

Distance: 0.012057721614837646



9. Image ID: 8640

Distance: 0.014498233795166016



10. Image ID: 8638

Distance: 0.015948235988616943

